

Quantum Information and Computation

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1 Quantum Theory Foundations

Quantum mechanics describes physical systems in terms of states, observables, measurements, and dynamical evolution. Unlike classical mechanics, where a physical state is represented by a point in phase space, a pure quantum state can be represented by a normalized state vector

$$|\psi\rangle \in \mathcal{H},$$

where $\mathcal{H}$ is the Hilbert space associated with the system.

The state vector satisfies the normalization condition

$$\langle\psi|\psi\rangle = 1.$$

The mathematical structure of $\mathcal{H}$ provides the framework for describing superposition, interference, measurement probabilities, and composite systems.

1.1 Hilbert Spaces and Dirac Notation

A Hilbert space is a vector space equipped with an inner product and complete with respect to the norm induced by that inner product. In quantum mechanics, the Hilbert space provides the mathematical setting in which quantum states are represented.

For a quantum system, the Hilbert space is a complex vector space. We denote it by

$$\mathcal{H}.$$

1.1.1 Dirac Notation

Dirac notation provides a convenient way to represent quantum states and the mathematical operations involving them.

A pure quantum state is represented by a state vector. In Dirac notation, a state vector is written as a *ket*:

$$|\psi\rangle.$$

1.1.2 Dual Vectors

For every ket $|\psi\rangle$ in a Hilbert space, there is a corresponding dual vector, called a *bra* and denoted by $\langle\psi|$. It is represented by the Hermitian adjoint of the ket:

$$\langle\psi| = |\psi\rangle^\dagger.$$

For example, if

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix},$$

then

$$\langle\psi| = (\alpha^* \quad \beta^*),$$

where $*$ denotes complex conjugation.

More generally,

$$(c|\psi\rangle)^\dagger = c^* \langle\psi|, \quad c \in \mathbb{C}.$$

The bra and ket can be combined to form an inner product:

$$\langle\phi||\psi\rangle.$$

1.1.3 Vector Addition and Scalar Multiplication

Kets obey the same linear structure as vectors. They can be added together and multiplied by complex scalars:

$$|\psi\rangle + |\phi\rangle = |\chi\rangle,$$

and

$$c|\psi\rangle = |\xi\rangle, \quad c \in \mathbb{C}.$$

These properties allow quantum states to be expressed as linear combinations of basis states.

1.2 Inner Products and Orthogonality

For two vectors $|\phi\rangle, |\psi\rangle \in \mathcal{H}$, their inner product is denoted by

$$\langle\phi|\psi\rangle.$$

For vectors in $\mathbb{C}^n$,

$$\langle\phi|\psi\rangle = \sum_{i=1}^n \phi_i^* \psi_i.$$

The inner product is a complex number and provides a way to measure the relationship between vectors in a Hilbert space.

Norm The norm of a vector $|\psi\rangle$ is defined by

$$\|\psi\| = \sqrt{\langle\psi|\psi\rangle}.$$

A vector is normalized when

$$\|\psi\| = 1,$$

or equivalently,

$$\langle\psi|\psi\rangle = 1.$$

Orthogonality

Two vectors $|\phi\rangle$ and $|\psi\rangle$ are orthogonal if

$$\langle\phi|\psi\rangle = 0.$$

Thus, orthogonality in a Hilbert space is the generalization of perpendicularity in Euclidean space.

Orthonormality

A set of vectors $\{|i\rangle\}$ is orthonormal if its vectors are normalized and mutually orthogonal. This condition can be written compactly using the *Kronecker delta*:

$$\boxed{\langle i|j\rangle = \delta_{ij}}$$

where

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Thus, δ_{ij} expresses both normalization and orthogonality:

$$\langle i|i\rangle = 1, \quad \langle i|j\rangle = 0 \quad (i \neq j).$$

Completeness

An orthonormal set is complete if it spans the Hilbert space. For a complete orthonormal basis $\{|i\rangle\}$, every vector $|\psi\rangle \in \mathcal{H}$ can be expanded as

$$|\psi\rangle = \sum_i c_i |i\rangle,$$

where

$$c_i = \langle i|\psi\rangle.$$

Equivalently, completeness is expressed by the relation

$$\boxed{\sum_i |i\rangle \langle i| = I}.$$

1.3 Classical and Quantum States

In classical physics, the state of a system is specified by the values of its physical variables. In the case of a classical bit, the state is one of two definite values,

$$0 \quad \text{or} \quad 1.$$

A quantum state, in contrast, is described by a state vector in a Hilbert space. For a single qubit, the computational basis states are denoted by

$$|0\rangle \quad \text{and} \quad |1\rangle.$$

Unlike a classical bit, a qubit can also exist in a superposition of these basis states.

1.3.1 Single-Qubit Systems

For a qubit, the Hilbert space is

$$\mathcal{H} = \mathbb{C}^2.$$

A commonly used basis is the computational basis, also called the Z -basis:

$$\mathcal{B}_Z = \{|0\rangle, |1\rangle\},$$

where

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The computational basis is a complete orthonormal basis for $\mathbb{C}^2$. Therefore,

$$\langle i | j \rangle = \delta_{ij}, \quad i, j \in \{0, 1\},$$

and

$$|0\rangle \langle 0| + |1\rangle \langle 1| = I.$$

The X -basis

Another complete orthonormal basis for a qubit is the X -basis:

$$\mathcal{B}_X = \{|+\rangle, |-\rangle\},$$

where

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle),$$

$$|- \rangle = \frac{1}{\sqrt{2}} (|0 \rangle - |1 \rangle) .$$

These states satisfy

$$\langle + | + \rangle = \langle - | - \rangle = 1 ,$$

and

$$\langle + | - \rangle = 0 .$$

They also satisfy the completeness relation

$$|+ \rangle \langle + | + |- \rangle \langle - | = I .$$

The Z -basis and X -basis are two different complete orthonormal bases for the same qubit Hilbert space. In general, the state of a qubit can be expressed in either basis.

The Y -basis

The Y -basis is the eigenbasis of the Pauli- Y operator:

$$\mathcal{B}_Y = \{ |+i \rangle , |-i \rangle \} ,$$

where

$$|+i \rangle = \frac{1}{\sqrt{2}} (|0 \rangle + i |1 \rangle) ,$$

$$|-i \rangle = \frac{1}{\sqrt{2}} (|0 \rangle - i |1 \rangle) .$$

These states form a complete orthonormal basis for $\mathbb{C}^2$:

$$\langle +i | +i \rangle = \langle -i | -i \rangle = 1 ,$$

$$\langle +i | -i \rangle = 0 ,$$

and

$$|+i \rangle \langle +i | + |-i \rangle \langle -i | = I .$$

Bloch Sphere

Any pure state of a single qubit can be written, up to an irrelevant global phase, as

$$|\psi \rangle = \cos \frac{\theta}{2} |0 \rangle + e^{i\phi} \sin \frac{\theta}{2} |1 \rangle ,$$

where

$$0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi.$$

The parameters θ and ϕ specify a point on the surface of the Bloch sphere. The corresponding Bloch vector is

$$\mathbf{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

For a pure state, the Bloch vector has unit length,

$$|\mathbf{r}| = 1,$$

so the state lies on the surface of the Bloch sphere.

The computational (Z -) basis states correspond to the two poles of the sphere:

$$|0\rangle \leftrightarrow (0, 0, 1), \quad |1\rangle \leftrightarrow (0, 0, -1).$$

The X -basis states correspond to the two directions of the x -axis:

$$|+\rangle \leftrightarrow (1, 0, 0), \quad |-\rangle \leftrightarrow (-1, 0, 0).$$

The Y -basis states correspond to the two directions of the y -axis:

$$|+i\rangle \leftrightarrow (0, 1, 0), \quad |-i\rangle \leftrightarrow (0, -1, 0).$$

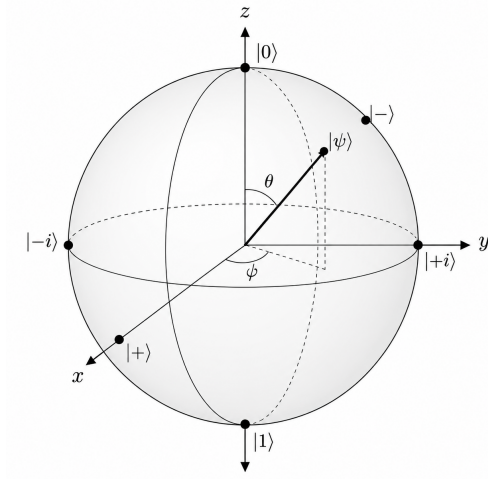


Figure 1: The Bloch sphere representation of a single-qubit pure state. The computational (Z -) basis states correspond to the $\pm z$ directions, the X -basis states to the $\pm x$ directions, and the Y -basis states to the $\pm y$ directions.

Thus, the X -, Y -, and Z -basis states correspond to the six points on the Bloch sphere along the positive and negative directions of the three mutually orthogonal axes.

1.3.2 Superposition

A general pure state of a single qubit can be written as a linear combination of the computational basis states:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle,$$

where $\alpha, \beta \in \mathbb{C}$ are called *probability amplitudes*.

Because the state vector is normalized,

$$\langle\psi|\psi\rangle = 1.$$

Using the orthonormality of the computational basis,

$$|\alpha|^2 + |\beta|^2 = 1.$$

The quantities $|\alpha|^2$ and $|\beta|^2$ determine the probabilities of obtaining the outcomes 0 and 1, respectively, when the qubit is measured in the computational basis.

Global Phase

Multiplying a quantum state by a phase factor $e^{i\gamma}$ gives

$$|\psi'\rangle = e^{i\gamma} |\psi\rangle.$$

The global phase $e^{i\gamma}$ has no observable physical effect. In particular, $|\psi\rangle$ and $e^{i\gamma} |\psi\rangle$ represent the same physical state.

Relative Phase

In a superposition

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle,$$

the relative phase between α and β can affect the outcome of subsequent quantum operations and measurements in other bases.

For example,

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

and

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

have the same computational-basis probabilities,

$$P(0) = P(1) = \frac{1}{2},$$

but differ in their relative phase.

Quantum Interference

Quantum interference arises because probability amplitudes, rather than probabilities, are added when different quantum alternatives contribute to the same outcome. The resulting probability is then obtained by taking the squared magnitude of the total amplitude.

Consider the two states

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

They have the same probabilities when measured in the computational basis:

$$P(0) = P(1) = \frac{1}{2}.$$

However, they have different relative phases. The relative sign between the two components affects how their amplitudes combine under subsequent quantum operations.

For example, applying a Hadamard gate gives

$$H|+\rangle = H\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |0\rangle,$$

whereas

$$H|-\rangle = H\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |1\rangle.$$

Thus, the two amplitudes combine constructively for one computational basis outcome and destructively for the other. The relative phase therefore changes the resulting measurement probabilities.

This constructive and destructive combination of probability amplitudes is the essence of quantum interference.

1.4 Operators and Outer Products

1.4.1 Operators

An operator is a linear map that acts on vectors in a Hilbert space. For an operator A acting on $\mathcal{H}$,

$$A : \mathcal{H} \rightarrow \mathcal{H}.$$

Its action on a state vector $|\psi\rangle$ produces another vector in the Hilbert space:

$$A|\psi\rangle = |\phi\rangle.$$

An operator is linear if

$$A(a|\psi\rangle + b|\phi\rangle) = aA|\psi\rangle + bA|\phi\rangle, \quad a, b \in \mathbb{C}.$$

Operators are used to describe physical quantities, transformations, and measurements in quantum mechanics.

1.4.2 Matrix Representation

Once a basis is chosen, an operator can be represented by a matrix. For a qubit in the computational basis $\{|0\rangle, |1\rangle\}$, a state vector has the form

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix},$$

and an operator can be represented by a 2×2 matrix,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The action of the operator is then ordinary matrix multiplication:

$$A|\psi\rangle = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

The matrix elements of an operator in an orthonormal basis $\{|i\rangle\}$ are given by

$$A_{ij} = \langle i | A | j \rangle.$$

For example, consider the Pauli- X operator

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

in the computational basis $\mathcal{B}_Z = \{|0\rangle, |1\rangle\}$.

Its matrix elements are

$$\begin{aligned} X_{00} &= \langle 0 | X | 0 \rangle = 0, \\ X_{01} &= \langle 0 | X | 1 \rangle = 1, \\ X_{10} &= \langle 1 | X | 0 \rangle = 1, \\ X_{11} &= \langle 1 | X | 1 \rangle = 0. \end{aligned}$$

Therefore,

$$X = \begin{pmatrix} X_{00} & X_{01} \\ X_{10} & X_{11} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Conversely, an operator can be reconstructed from its matrix elements. For an orthonormal basis $\{|i\rangle\}$,

$$A = \sum_{i,j} A_{ij} |i\rangle \langle j|.$$

For the Pauli- X operator, this gives

$$X = X_{00} |0\rangle \langle 0| + X_{01} |0\rangle \langle 1| + X_{10} |1\rangle \langle 0| + X_{11} |1\rangle \langle 1|.$$

Using the matrix elements above,

$$X = |0\rangle \langle 1| + |1\rangle \langle 0|.$$

Indeed,

$$|0\rangle \langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad |1\rangle \langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

so that

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The matrix representation of an operator depends on the choice of basis. For example, consider the Pauli- Z operator

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

in the computational basis $\mathcal{B}_Z = \{|0\rangle, |1\rangle\}$.

In the X -basis

$$\mathcal{B}_X = \{|+\rangle, |-\rangle\},$$

the same operator has the matrix representation

$$[Z]_{\mathcal{B}_X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Thus, the physical operator is unchanged, but its matrix representation changes when the basis changes.

1.4.3 Outer Product

An outer product combines a ket and a bra to form an operator:

$$|\psi\rangle \langle \phi|.$$

This should be distinguished from the inner product

$$\langle \phi | \psi \rangle,$$

which is a complex number.

In contrast,

$$|\psi\rangle\langle\phi|$$

is an operator.

For example,

$$|0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Similarly,

$$|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

An operator of the form

$$P = |\psi\rangle\langle\psi|$$

is a projector when $|\psi\rangle$ is normalized. It satisfies

$$P^2 = P.$$

For an orthonormal basis $\{|i\rangle\}$, the operators

$$P_i = |i\rangle\langle i|$$

are the corresponding basis projectors.

Example: Projecting onto a Basis State

Consider the projector onto the state $|0\rangle$:

$$P_0 = |0\rangle \langle 0|.$$

For a general qubit state

$$|\phi\rangle = \alpha |0\rangle + \beta |1\rangle,$$

the action of P_0 is

$$\begin{aligned} P_0 |\phi\rangle &= |0\rangle \langle 0| (\alpha |0\rangle + \beta |1\rangle) \\ &= \alpha |0\rangle \langle 0|0\rangle + \beta |0\rangle \langle 0|1\rangle \\ &= \alpha |0\rangle. \end{aligned}$$

Thus, P_0 extracts the component of $|\phi\rangle$ along $|0\rangle$.

Projector Property

Applying the projector twice gives

$$\begin{aligned} P_0^2 &= (|0\rangle \langle 0|)(|0\rangle \langle 0|) \\ &= |0\rangle \underbrace{\langle 0|0\rangle}_{=1} \langle 0| \\ &= |0\rangle \langle 0| \\ &= P_0. \end{aligned}$$

Therefore,

$$\boxed{P_0^2 = P_0}.$$

This means that applying the same projector twice has the same effect as applying it once.

1.4.4 Hermitian Operators and Observables

The Hermitian adjoint of an operator A is denoted by $A^\dagger$. An operator is Hermitian if

$$A^\dagger = A.$$

Hermitian operators have real eigenvalues. In quantum mechanics, observables are represented by Hermitian operators.

For example, the Pauli-Z operator is

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and it is Hermitian because

$$Z^\dagger = Z.$$

The expectation value of an observable A in a pure state $|\psi\rangle$ is

$$\langle A \rangle_\psi = \langle \psi | A | \psi \rangle .$$

Example: Pauli- Z as a Hermitian Operator

Consider the Pauli- Z operator

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} .$$

Its Hermitian adjoint is

$$Z^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z ,$$

so Z is Hermitian.

The eigenvalue equation

$$Z |\lambda\rangle = \lambda |\lambda\rangle$$

gives

$$Z |0\rangle = |0\rangle , \quad Z |1\rangle = -|1\rangle .$$

Therefore, the eigenvalues of Z are

$$\lambda_0 = +1, \quad \lambda_1 = -1,$$

which are real, as expected for a Hermitian operator.

1.4.5 Eigenvalues and Eigenvectors

An eigenvector of an operator A is a nonzero vector that satisfies

$$A|\lambda\rangle = \lambda|\lambda\rangle,$$

where λ is the corresponding eigenvalue.

For example, the computational basis states are eigenvectors of the Pauli- Z operator:

$$Z|0\rangle = +|0\rangle,$$

$$Z|1\rangle = -|1\rangle.$$

Thus, the states $|0\rangle$ and $|1\rangle$ are the eigenstates of Z with eigenvalues $+1$ and -1 , respectively.

More generally, for a Hermitian observable, the eigenvalues are real and the eigenvectors corresponding to distinct eigenvalues are orthogonal. The eigenvectors therefore provide a natural basis for describing measurements of the observable.

1.5 Quantum Measurement

Quantum measurement is the process by which information about a quantum system is extracted. A measurement is associated with a set of possible outcomes, and the outcome obtained in a particular experiment is probabilistic.

For a projective measurement in an orthonormal basis $\{|i\rangle\}$, the corresponding projectors are

$$P_i = |i\rangle\langle i|.$$

These projectors satisfy

$$P_i^\dagger = P_i, \quad P_i^2 = P_i,$$

and, for a complete orthonormal basis,

$$\sum_i P_i = I.$$

The projectors identify the possible measurement outcomes and are used to calculate their probabilities.

1.5.1 The Born Rule

When a pure quantum state is expressed in an orthonormal measurement basis $\{|i\rangle\}$,

$$|\psi\rangle = \sum_i c_i |i\rangle,$$

the coefficient c_i is the probability amplitude associated with the outcome i .
The Born rule states that the probability of obtaining outcome i is

$$\boxed{P(i) = |c_i|^2}.$$

Equivalently, using the projector $P_i = |i\rangle \langle i|$,

$$P(i) = \langle \psi | P_i | \psi \rangle = | \langle i | \psi \rangle |^2.$$

Example: Measurement of a qubit

Consider the quantum state

$$|\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle.$$

For a measurement in the computational basis $\{|0\rangle, |1\rangle\}$, the probability amplitudes are

$$c_0 = \frac{1}{\sqrt{2}}, \quad c_1 = \frac{1}{\sqrt{2}}.$$

Applying the Born rule,

$$P(0) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2},$$

and

$$P(1) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

Therefore, measuring this state in the computational basis gives

$$\boxed{P(0) = 50\%}$$

and

$$\boxed{P(1) = 50\%}.$$

Because these are the only possible outcomes of the measurement,

$$P(0) + P(1) = 1.$$

For a normalized state,

$$\langle \psi | \psi \rangle = 1,$$

and, for an orthonormal basis,

$$\langle \psi | \psi \rangle = \sum_i |c_i|^2.$$

Therefore,

$$\boxed{\sum_i P(i) = 1}.$$

Thus, normalization ensures that the probabilities obtained from the Born rule form a valid probability distribution.

1.5.2 Post-Measurement State

If measurement outcome i is obtained, the state is projected onto the subspace associated with that outcome. For a projective measurement, the post-measurement state is

$$\boxed{|\psi_i\rangle = \frac{P_i |\psi\rangle}{\sqrt{P(i)}}$$

provided that $P(i) > 0$.

For example, consider

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

and the projector

$$P_0 = |0\rangle \langle 0|.$$

Then

$$P_0 |\psi\rangle = \alpha |0\rangle.$$

The probability of obtaining outcome 0 is

$$P(0) = |\alpha|^2.$$

Therefore, after obtaining outcome 0, the normalized post-measurement state is

$$|\psi_0\rangle = \frac{\alpha |0\rangle}{|\alpha|}$$

which differs from $|0\rangle$ only by a global phase. Thus, physically, the post-measurement state is $|0\rangle$.

1.5.3 Measurement Basis

The probabilities obtained in a measurement depend on the basis in which the measurement is performed. The same quantum state can therefore produce different probability distributions when measured in different bases.

For example, the computational (Z -) basis is

$$\mathcal{B}_Z = \{|0\rangle, |1\rangle\},$$

while the X -basis is

$$\mathcal{B}_X = \{|+\rangle, |-\rangle\},$$

where

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

Thus, specifying a quantum measurement requires specifying the measurement being performed, including its associated basis or projectors.

1.6 Unitary Evolution

Unitary Operators

The time evolution of an isolated quantum system is described by a unitary operator U . A unitary operator satisfies

$$U^\dagger U = U U^\dagger = I.$$

If the initial state is $|\psi\rangle$, its state after the evolution is

$$|\psi'\rangle = U |\psi\rangle.$$

Unitarity preserves inner products and therefore preserves the norm of quantum states:

$$\langle\psi'|\psi'\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle.$$

Thus, if $|\psi\rangle$ is normalized, $U |\psi\rangle$ is also normalized.

Quantum Gates

In quantum computing, a quantum gate is a unitary operation acting on one or more qubits. A gate transforms an input state into an output state according to

$$|\psi_{\text{out}}\rangle = U |\psi_{\text{in}}\rangle.$$

Because quantum gates are unitary, they preserve the normalization of quantum states and are reversible.

Pauli Gates

The Pauli operators are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are Hermitian and unitary:

$$X^\dagger = X, \quad Y^\dagger = Y, \quad Z^\dagger = Z,$$

and

$$X^\dagger X = Y^\dagger Y = Z^\dagger Z = I.$$

Their action on the computational basis is

$$X |0\rangle = |1\rangle, \quad X |1\rangle = |0\rangle,$$

$$Y |0\rangle = i |1\rangle, \quad Y |1\rangle = -i |0\rangle,$$

$$Z |0\rangle = |0\rangle, \quad Z |1\rangle = -|1\rangle.$$

Hadamard Gate

The Hadamard gate is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its action on the computational basis states is

$$H |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle,$$

and

$$H |1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle.$$

The Hadamard gate therefore transforms the computational basis into the X -basis and vice versa:

$$H |+\rangle = |0\rangle, \quad H |-\rangle = |1\rangle.$$

Phase Gates

A phase gate changes the relative phase between the computational basis states without changing their measurement probabilities in the computational basis.

The general single-qubit phase gate is

$$P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}.$$

Its action is

$$P(\phi) |0\rangle = |0\rangle,$$

$$P(\phi) |1\rangle = e^{i\phi} |1\rangle.$$

Important special cases include the S and T gates:

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

Controlled Gates

A controlled gate applies an operation to a target qubit conditional on the state of a control qubit. The most common example is the controlled-NOT (CNOT) gate. Here, the first qubit is the control qubit and the second qubit is the target qubit.

Its action on the computational basis is

$$|00\rangle \rightarrow |00\rangle,$$

$$|01\rangle \rightarrow |01\rangle,$$

$$|10\rangle \rightarrow |11\rangle,$$

$$|11\rangle \rightarrow |10\rangle.$$

Its matrix representation, in the ordered basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, is

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The CNOT gate can create entanglement when acting on an appropriate superposition state. For example, consider the product state

$$|+0\rangle = |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle).$$

Applying CNOT gives

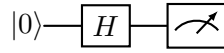
$$\begin{aligned}\text{CNOT } | +0 \rangle &= \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \\ &= |\Phi^+\rangle.\end{aligned}$$

Thus, the CNOT gate can transform a separable input state into an entangled Bell state.

Quantum Circuit Representation

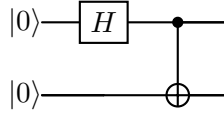
Quantum circuits provide a graphical representation of quantum computation. Horizontal lines represent qubits, while gates are represented by operations placed on those lines. Time flows from left to right.

For example, the circuit



represents a qubit initially in $|0\rangle$, followed by a Hadamard gate and a measurement.

For a two-qubit circuit,



the Hadamard gate creates a superposition, and the CNOT gate acts conditionally on the first qubit.

1.7 Composite Quantum Systems

Tensor Products When a quantum system consists of two subsystems A and B , the Hilbert space of the composite system is the tensor product of the Hilbert spaces of the individual subsystems:

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B.$$

Here, $\mathcal{H}_A$ and $\mathcal{H}_B$ are the Hilbert spaces of subsystems A and B , respectively.

If $\{|a_i\rangle_A\}$ is an orthonormal basis for $\mathcal{H}_A$ and $\{|b_j\rangle_B\}$ is an orthonormal basis for $\mathcal{H}_B$, then the tensor products

$$|a_i\rangle_A \otimes |b_j\rangle_B$$

form an orthonormal basis for the composite Hilbert space $\mathcal{H}_{AB}$.

The dimension of the composite Hilbert space is

$$\dim(\mathcal{H}_{AB}) = \dim(\mathcal{H}_A) \dim(\mathcal{H}_B).$$

Two-Qubit System

For two qubits,

$$\mathcal{H}_A = \mathbb{C}^2, \quad \mathcal{H}_B = \mathbb{C}^2,$$

so that

$$\mathcal{H}_{AB} = \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4.$$

The computational bases of the two qubits are

$$\mathcal{B}_A = \{|0\rangle_A, |1\rangle_A\}, \quad \mathcal{B}_B = \{|0\rangle_B, |1\rangle_B\}.$$

The corresponding computational basis of the composite system is

$$\mathcal{B}_{AB} = \{|0\rangle_A \otimes |0\rangle_B, |0\rangle_A \otimes |1\rangle_B, |1\rangle_A \otimes |0\rangle_B, |1\rangle_A \otimes |1\rangle_B\}.$$

It is customary to write these states more compactly as

$$|00\rangle, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

Tensor-Product Basis States

Using the column-vector representation,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

the tensor products give

$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

$$|01\rangle = |0\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

$$|10\rangle = |1\rangle \otimes |0\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix},$$

and

$$|11\rangle = |1\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Thus, the four tensor-product basis states form the standard basis of the four-dimensional Hilbert space of two qubits.

General Bipartite State

A general pure state of the composite system can be expressed in the product basis as

$$|\psi\rangle_{AB} = \sum_{i,j} c_{ij} |a_i\rangle_A \otimes |b_j\rangle_B,$$

where $c_{ij} \in \mathbb{C}$ are the probability amplitudes.

Equivalently,

$$|\psi\rangle_{AB} = \sum_{i,j} c_{ij} |a_i b_j\rangle.$$

For a normalized state,

$$\langle\psi|\psi\rangle = \sum_{i,j} |c_{ij}|^2 = 1.$$

A general bipartite state need not be expressible as a single tensor product of a state of A and a state of B . This distinction leads naturally to the concept of entanglement.

More Than Two Subsystems

The tensor-product construction extends naturally to systems containing more than two subsystems. For three subsystems A , B , and C ,

$$\mathcal{H}_{ABC} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C.$$

For three qubits,

$$\mathcal{H}_{ABC} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^8.$$

More generally, an n -qubit system has Hilbert-space dimension

$$\boxed{\dim(\mathcal{H}) = 2^n}.$$

Thus, the number of computational-basis states of an n -qubit system is also 2^n . These basis states can be written as

$$|x_1 x_2 \cdots x_n\rangle, \quad x_k \in \{0, 1\}.$$

1.8 Entanglement

Separable and Entangled States

Consider a bipartite pure state $|\Psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$. The state $|\Psi\rangle_{AB}$ is said to be *separable* if it can be written as

$$|\Psi\rangle_{AB} = |\Phi\rangle_A \otimes |\varphi\rangle_B,$$

where

$$|\Phi\rangle_A \in \mathcal{H}_A, \quad |\varphi\rangle_B \in \mathcal{H}_B.$$

If no such decomposition exists, the state is called *entangled* or *nonseparable*:

$$|\Psi\rangle_{AB} \neq |\Phi\rangle_A \otimes |\varphi\rangle_B$$

for all $|\Phi\rangle_A \in \mathcal{H}_A$ and $|\varphi\rangle_B \in \mathcal{H}_B$.

Thus, for a bipartite pure state, entanglement is characterized by the inability to factorize the state into a tensor product of states of the individual subsystems.

Entanglement is a fundamental resource in quantum information. It enables correlations that have no classical analogue and plays a central role in quantum communication, quantum computation, and quantum information processing.

Bell States

The Bell states are four maximally entangled two-qubit states:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle),$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle),$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle),$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle).$$

The four Bell states form a complete orthonormal basis for the two-qubit Hilbert space. This basis is called the *Bell basis*.

The computational basis can be expressed in terms of the Bell basis:

$$|00\rangle = \frac{1}{\sqrt{2}} (|\Phi^+\rangle + |\Phi^-\rangle),$$

$$|01\rangle = \frac{1}{\sqrt{2}} (|\Psi^+\rangle + |\Psi^-\rangle),$$

$$|10\rangle = \frac{1}{\sqrt{2}} (|\Psi^+\rangle - |\Psi^-\rangle),$$

$$|11\rangle = \frac{1}{\sqrt{2}} (|\Phi^+\rangle - |\Phi^-\rangle).$$

Example: Demonstrating Entanglement

Suppose, for contradiction, that the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

can be written as a product state:

$$|\Phi^+\rangle = |\phi\rangle_A \otimes |\chi\rangle_B,$$

where

$$|\phi\rangle_A = a|0\rangle + b|1\rangle, \quad |\chi\rangle_B = c|0\rangle + d|1\rangle.$$

Their tensor product is

$$\begin{aligned} |\phi\rangle_A \otimes |\chi\rangle_B &= (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) \\ &= ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle. \end{aligned}$$

For this to equal

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle,$$

we would require

$$ac = \frac{1}{\sqrt{2}}, \quad ad = 0, \quad bc = 0, \quad bd = \frac{1}{\sqrt{2}}.$$

The first condition requires $a \neq 0$ and $c \neq 0$. Therefore, $ad = 0$ requires $d = 0$, while $bc = 0$ requires $b = 0$. But then

$$bd = 0,$$

which contradicts

$$bd = \frac{1}{\sqrt{2}}.$$

Therefore, $|\Phi^+\rangle$ cannot be written as a product state and is entangled.

2 Quantum Computation and Applications

2.1 Quantum Circuits

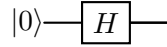
A quantum circuit is a sequence of quantum operations applied to one or more qubits. It provides a graphical representation of a quantum computation, where horizontal lines represent qubits and gates represent quantum operations. The circuit is read from left to right, with the left side representing the initial state and the right side the final state.

Single-Qubit Circuit

Consider a qubit initially prepared in $|0\rangle$. Applying a Hadamard gate produces the state

$$|0\rangle \xrightarrow{H} |+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle).$$

The corresponding circuit is



Multi-Qubit Circuit

Quantum circuits can contain gates acting on one or more qubits. For example, consider two qubits initially in the state $|00\rangle$. A Hadamard gate is applied to the first qubit, followed by a controlled-NOT (CNOT) gate with the first qubit as the control and the second qubit as the target.

The Hadamard operation on the first qubit, while leaving the second qubit unchanged, is represented by

$$H \otimes I.$$

In general, for product states,

$$(A \otimes B)(|\psi\rangle \otimes |\phi\rangle) = (A|\psi\rangle) \otimes (B|\phi\rangle).$$

Thus,

$$(H \otimes I)|00\rangle = (H|0\rangle) \otimes (I|0\rangle).$$

Since

$$H|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle), \quad I|0\rangle = |0\rangle,$$

we obtain

$$\begin{aligned} (H \otimes I)|00\rangle &= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |0\rangle \\ &= \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle). \end{aligned}$$

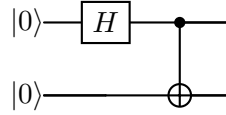
The CNOT gate flips the target qubit when the control qubit is in $|1\rangle$. Therefore,

$$\frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle).$$

The final state is the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle).$$

The corresponding circuit is



This circuit prepares the Bell state $|\Phi^+\rangle$ from the initial state $|00\rangle$.

Measurement in a Quantum Circuit

A measurement converts quantum information into a classical outcome. For example, measuring the state

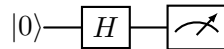
$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

in the computational basis gives

$$P(0) = \frac{1}{2}, \quad P(1) = \frac{1}{2}.$$

The measurement result is stored in a classical bit.

The corresponding circuit is



Classical Bits and Measurement Outcomes

A quantum circuit may contain classical bits used to store measurement outcomes. These classical results can subsequently be used for classical processing or to condition later quantum operations.

Gate Sequence and Circuit Depth

The order of operations in a quantum circuit is important because quantum operators do not generally commute. A circuit therefore specifies an ordered sequence of operations.

The *circuit depth* is the number of sequential layers of operations required to execute the circuit, where operations acting on different qubits can be performed in parallel when allowed by the hardware or circuit structure.

2.2 Quantum Teleportation

Suppose Alice has an unknown qubit state

$$|\psi\rangle_{A'} = \alpha |0\rangle_{A'} + \beta |1\rangle_{A'}, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (1)$$

Alice and Bob share a pre-established entangled Bell state,

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}} (|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B). \quad (2)$$

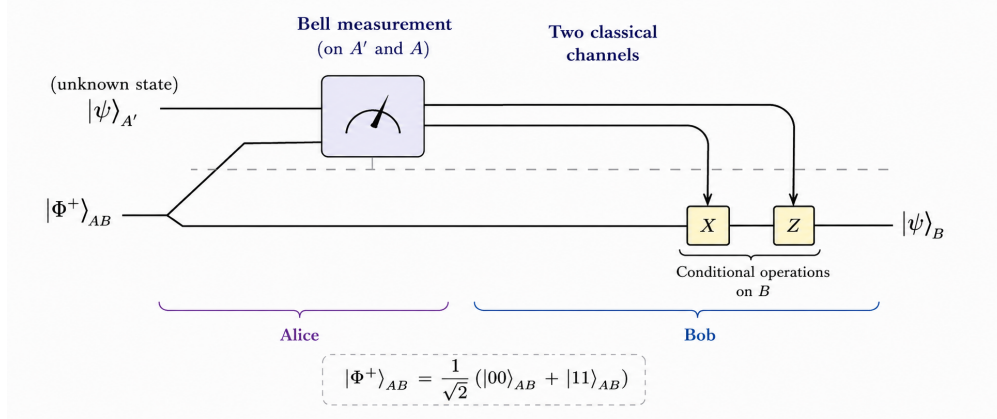


Figure 2: Schematic representation of the quantum teleportation protocol. Alice performs a Bell-basis measurement on the input qubit A' and her qubit A of a shared Bell pair. The two classical measurement bits are communicated to Bob, who applies the corresponding conditional Pauli operation to his qubit B , thereby obtaining the input state $|\psi\rangle_B$.

The complete state of the three qubits, ordered as $A'AB$, is therefore

$$\begin{aligned} |\varphi\rangle_{A'AB} &= |\psi\rangle_{A'} \otimes |\Phi^+\rangle_{AB} \\ &= (\alpha |0\rangle_{A'} + \beta |1\rangle_{A'}) \otimes \frac{1}{\sqrt{2}} (|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B). \end{aligned} \quad (3)$$

Expanding the tensor product gives

$$\begin{aligned} |\varphi\rangle_{A'AB} &= \frac{1}{\sqrt{2}} \left(\alpha |0\rangle_{A'} |0\rangle_A |0\rangle_B + \alpha |0\rangle_{A'} |1\rangle_A |1\rangle_B \right. \\ &\quad \left. + \beta |1\rangle_{A'} |0\rangle_A |0\rangle_B + \beta |1\rangle_{A'} |1\rangle_A |1\rangle_B \right). \end{aligned} \quad (4)$$

Alice will perform a joint measurement on her two qubits, A' and A , in the Bell basis. It is therefore useful to rewrite the state of these two qubits in terms of the Bell states.

$$\begin{aligned}
|00\rangle_{A'A} &= \frac{1}{\sqrt{2}} (|\Phi^+\rangle_{A'A} + |\Phi^-\rangle_{A'A}), \\
|11\rangle_{A'A} &= \frac{1}{\sqrt{2}} (|\Phi^+\rangle_{A'A} - |\Phi^-\rangle_{A'A}), \\
|01\rangle_{A'A} &= \frac{1}{\sqrt{2}} (|\Psi^+\rangle_{A'A} + |\Psi^-\rangle_{A'A}), \\
|10\rangle_{A'A} &= \frac{1}{\sqrt{2}} (|\Psi^+\rangle_{A'A} - |\Psi^-\rangle_{A'A}).
\end{aligned} \tag{5}$$

We now group Alice's two qubits together and keep Bob's qubit separate:

$$\begin{aligned}
|\varphi\rangle_{A'AB} &= \frac{1}{\sqrt{2}} \left[\alpha |00\rangle_{A'A} |0\rangle_B + \alpha |01\rangle_{A'A} |1\rangle_B \right. \\
&\quad \left. + \beta |10\rangle_{A'A} |0\rangle_B + \beta |11\rangle_{A'A} |1\rangle_B \right].
\end{aligned} \tag{6}$$

Substituting the Bell-basis expansions above and collecting terms according to the Bell state of $A'A$, we obtain

$$\begin{aligned}
|\varphi\rangle_{A'AB} &= \frac{1}{2} \left[|\Phi^+\rangle_{A'A} \otimes (\alpha |0\rangle_B + \beta |1\rangle_B) \right. \\
&\quad + |\Phi^-\rangle_{A'A} \otimes (\alpha |0\rangle_B - \beta |1\rangle_B) \\
&\quad + |\Psi^+\rangle_{A'A} \otimes (\beta |0\rangle_B + \alpha |1\rangle_B) \\
&\quad \left. + |\Psi^-\rangle_{A'A} \otimes (-\beta |0\rangle_B + \alpha |1\rangle_B) \right].
\end{aligned} \tag{7}$$

Equation (7) shows that, conditional on the outcome of Alice's Bell-basis measurement, Bob's qubit is projected onto one of four possible states:

$$\begin{aligned}
|\Phi^+\rangle_{A'A} &\longrightarrow \alpha |0\rangle_B + \beta |1\rangle_B, \\
|\Phi^-\rangle_{A'A} &\longrightarrow \alpha |0\rangle_B - \beta |1\rangle_B, \\
|\Psi^+\rangle_{A'A} &\longrightarrow \beta |0\rangle_B + \alpha |1\rangle_B, \\
|\Psi^-\rangle_{A'A} &\longrightarrow -\beta |0\rangle_B + \alpha |1\rangle_B.
\end{aligned} \tag{8}$$

These four conditional states can be expressed in terms of the Pauli operators as

$$\begin{aligned}
|\Phi^+\rangle_{A'A} &\longrightarrow I |\psi\rangle_B, \\
|\Phi^-\rangle_{A'A} &\longrightarrow Z |\psi\rangle_B, \\
|\Psi^+\rangle_{A'A} &\longrightarrow X |\psi\rangle_B, \\
|\Psi^-\rangle_{A'A} &\longrightarrow XZ |\psi\rangle_B.
\end{aligned} \tag{9}$$

Since the Bell states form an orthonormal basis and each conditional state of Bob's qubit is normalized, each branch in Eq. (7) has probability

$$p = \left| \frac{1}{2} \right|^2 = \frac{1}{4}. \tag{10}$$

Table 1: Bell-measurement outcomes, conditional states of Bob’s qubit, and the corresponding Pauli corrections.

Alice’s outcome	Bob’s conditional state	Bob’s correction
$ \Phi^+\rangle_{A'A}$	$I \psi\rangle_B$	I
$ \Phi^-\rangle_{A'A}$	$Z \psi\rangle_B$	Z
$ \Psi^+\rangle_{A'A}$	$X \psi\rangle_B$	X
$ \Psi^-\rangle_{A'A}$	$XZ \psi\rangle_B$	XZ

Alice communicates the outcome of her Bell-basis measurement to Bob using two classical bits. These bits identify which of the four Bell states was obtained and therefore determine the corresponding Pauli correction that Bob must apply to his qubit. After applying the appropriate correction, Bob obtains the original input state:

$$\boxed{|\psi\rangle_B = \alpha |0\rangle_B + \beta |1\rangle_B}. \quad (11)$$

2.3 Quantum Algorithms

2.3.1 Quantum Parallelism and Phase Kickback

Quantum Parallelism and Phase Kickback

Quantum parallelism refers to the ability of a quantum computer to apply a unitary operation to a superposition of computational-basis states.

For a composite system, tensor-product states are often written in a compact form. For example,

$$|x, y\rangle \equiv |x\rangle \otimes |y\rangle,$$

where x and y label the states of two separate quantum registers.

An oracle U_f for a Boolean function f acts according to

$$U_f |x, y\rangle = |x, y \oplus f(x)\rangle,$$

The first register is the input (or data) register, while the second register is the target register.

For example, if the first register is prepared in the state

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle),$$

then, by linearity,

$$\begin{aligned} U_f |+, y\rangle &= U_f \left[\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |y\rangle \right] \\ &= \frac{1}{\sqrt{2}} (|0\rangle |y \oplus f(0)\rangle + |1\rangle |y \oplus f(1)\rangle). \end{aligned}$$

Thus, the oracle acts on both components of the superposition in a single application.

Phase Kickback

Phase kickback occurs when a controlled operation acts on a target system prepared in an eigenstate of the operation applied to that system. The corresponding eigenvalue appears as a phase factor on the control system.

Consider the state

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

The state $|-\rangle$ is an eigenstate of the Pauli- X operator:

$$X|-\rangle = -|-\rangle.$$

For a CNOT gate, the Pauli- X operation is applied to the target qubit when the control qubit is in $|1\rangle$. Therefore,

$$\text{CNOT } |0\rangle |-\rangle = |0\rangle |-\rangle,$$

while

$$\text{CNOT } |1\rangle |-\rangle = -|1\rangle |-\rangle.$$

Thus, for $b \in \{0, 1\}$,

$$\boxed{\text{CNOT } |b\rangle |-\rangle = (-1)^b |b\rangle |-\rangle}.$$

For a superposition of control states,

$$\text{CNOT } (\alpha_0 |0\rangle + \alpha_1 |1\rangle) |-\rangle = (\alpha_0 |0\rangle - \alpha_1 |1\rangle) |-\rangle.$$

The target state remains $|-\rangle$, while the eigenvalue -1 appears as a phase factor on the $|1\rangle$ component of the control state.

General Oracle

The same mechanism applies to an oracle

$$U_f |x, y\rangle = |x, y \oplus f(x)\rangle,$$

where $\oplus$ denotes addition modulo 2. With the target qubit prepared in $|-\rangle$,

$$\begin{aligned} U_f |x, -\rangle &= \frac{1}{\sqrt{2}} (|x, f(x)\rangle - |x, 1 \oplus f(x)\rangle) \\ &= (-1)^{f(x)} |x, -\rangle. \end{aligned}$$

Therefore,

$$\boxed{U_f |x, -\rangle = (-1)^{f(x)} |x, -\rangle}.$$

Since $f(x) \in \{0, 1\}$,

$$f(x) = 0 \Rightarrow (-1)^{f(x)} = +1, \quad f(x) = 1 \Rightarrow (-1)^{f(x)} = -1.$$

Thus, the value of $f(x)$ is encoded in a phase factor on the input basis state. When the input is a superposition of basis states, these phase factors can produce relative phases that are converted into measurable probability differences through quantum interference.

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